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SEMICONDUCTOR PACKAGES HAVING CONDUCTIVE PATTERNS OF REDISTRIBUTION STRUCTURE HAVING ELLIPSE-LIKE SHAPE

Abstract

A semiconductor package includes a die and a redistribution structure. The die has a plurality of corner regions and a first region between the corner regions. The redistribution structure is electrically connected to the die and includes a plurality of first conductive patterns disposed in the corner regions and a plurality of second conductive patterns disposed in the first region. The plurality of first conductive patterns have an ellipse-like shape respectively, and the plurality of second conductive patterns have a circular shape respectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 18/359,898, filed on Jul. 27, 2023. The prior application Ser. No. 18/359,898 is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/458,560, filed on Aug. 27, 2021. The entirety of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

[0002] The semiconductor industry has experienced rapid growth due to continuous improvements in the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, this improvement in integration density has come from repeated reductions in minimum feature size, which allows more of the smaller components to be integrated into a given area. These smaller electronic components also require smaller packages that utilize less area than previous packages. Currently, integrated fan-out packages are becoming increasingly popular for their compactness. How to ensure the reliability of the integrated fan-out packages has become a challenge in the field.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1A to FIG. 1E are schematic cross-sectional views illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.

[0004] FIG. 2A and FIG. 2B are simplified top views illustrating various arrangements of the conductive patterns in FIG. 1E.

[0005] FIG. 3A is a schematic cross-sectional view of a semiconductor package in accordance with some embodiments of the disclosure, and FIG. 3B is a simplified top view of FIG. 3A.

[0006] FIG. 4A and FIG. 4B are simplified top views illustrating various arrangements of the conductive patterns in FIG. 1E.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a second feature over or on a first feature in the description that follows may include embodiments in which the second and first features are formed in direct contact, and may also include embodiments in which

additional features may be formed between the second and first features, such that the second and first features may not be in direct contact. In addition, the disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “top,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0009] Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

[0010] FIG. **1A** to FIG. **1E** are schematic cross-sectional views illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure. Referring to FIG. **1A**, a carrier **102** having a de-bonding layer **104** thereon is provided. In some embodiments, the de-bonding layer **104** is formed on a top surface of the carrier **102**. For example, the carrier **102** is a glass substrate and the de-bonding layer **104** is a light-to-heat conversion (LTHC) release layer formed on the glass substrate. However, the disclosure is not limited thereto, and other suitable materials may be adapted for the carrier **102** and the de-bonding layer **104**. In alternative embodiments, a buffer layer (not shown) is coated on the de-bonding layer **104**, where the de-bonding layer **104** is sandwiched between the buffer layer and the carrier **102**, and a top surface of the buffer layer further provides a high degree of coplanarity. The buffer layer may be a dielectric material layer or a polymer layer which is made of polyimide, BCB, PBO, or any other suitable polymer-based dielectric material.

[0011] Then, a redistribution structure **110** is formed over the carrier **102**. In some embodiments, first, a passivation layer **112** is formed on the de-bonding layer **104**. A material of the passivation layer **112** may include polyimide, polyimide derivative, polybenzoxazole (PBO), or any other suitable dielectric material. The passivation layer **112** may be formed by spin-on coating, chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), or the like. The passivation layer **112** is a single layer or multiple layers. After forming the passivation layer **112**, a plurality of conductive vias **114** are formed in the passivation layer **112**, and a plurality of conductive patterns **116** are formed on the conductive vias **114** to electrically connect to the conductive vias **114** respectively. In some embodiments, a plurality of openings (not shown) are formed in the passivation layer **112**, and the conductive vias **114** are formed in the openings. The conductive patterns **116** are then formed on the conductive vias **114** respectively. In some embodiments, the conductive vias **114** and the conductive patterns **116** are formed integrally by a dual damascene process. In alternative embodiments, the conductive vias **114** and the conductive patterns **116** are formed separately by a single damascene process or any other suitable process. The conductive vias **114** and the conductive patterns **116** may each include a diffusion barrier layer and a conductive material thereon. The diffusion barrier layer may include one or more layers of TaN, Ta, TiN, Ti, CoW, or the like and be formed by ALD, or the like, and the conductive material may include copper, aluminum, tungsten, silver, combinations thereof, or the like and be formed by

CVD, PVD, a plating process, or the like. The number of the conductive vias **114** and the conductive patterns **116** is not limited in the disclosure.

[0012] After forming the conductive patterns **116**, a plurality of dielectric layers **122** and a plurality of conductive patterns **124** are alternately formed over the conductive patterns **116**, to complete the formation of the redistribution structure **110**. The conductive patterns **124** may include vias and/or traces to interconnect any devices and/or to an external device. The dielectric layers **122** may include silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, low-K dielectric material, such as PSG, BPSG, FSG, SiO.sub.xC.sub.y, Spin-On-Glass, Spin-On-Polymers, silicon carbon material, compounds thereof, composites thereof, combinations thereof, or the like. The dielectric layers **122** may be formed by spin-on coating, CVD, PECVD, HDP-CVD, or the like. The conductive patterns **124** are formed in/on the dielectric layers **122**. The conductive patterns **124** may include a diffusion barrier layer and a conductive material thereon. The diffusion barrier layer may include one or more layers of TaN, Ta, TiN, Ti, CoW, or the like and be formed by ALD, or the like, and the conductive material may include copper, aluminum, tungsten, silver, and combinations thereof, or the like and be formed by CVD, PVD, a plating process, or the like. The number of the dielectric layers **122** and the conductive patterns **124** is not limited in the disclosure.

[0013] Referring to FIG. **1B**, at least one die **130**, **140** is bonded to the redistribution structure **110**. The dies **130**, **140** may each have a single function (e.g., a logic device, memory die, etc.), or may have multiple functions (e.g., a system on a chip (SoC), an application-specific integrated circuit (ASIC), etc.). In some embodiments, the die **130** is a SoC, and the die **140** is a memory device. The memory device may be a dynamic random access memory (DRAM) die, a static random access memory (SRAM) die, a hybrid memory cube (HMC) module, a high bandwidth memory (HBM) module, or the like. In an embodiment, the die **140** is an HBM module.

[0014] In some embodiments, the dies **130**, **140** are bonded to the redistribution structure **110** with die connectors **134**, **144**. The die connectors **134**, **144** may be electrically connected to bond pads **132**, **142** of the dies **130**, **140**. In some embodiments, the dies **130**, **140** are placed on the redistribution structure **110** using a pick-and-place tool. The die connectors **134**, **144** may be formed from a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or combinations thereof. In some embodiments, the die connectors **134**, **144** are formed by initially forming a layer of solder through methods such as evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the die connectors **134**, **144** into desired bump shapes. The die connectors **134**, **144** form joints between corresponding conductive patterns **124** (i.e., the outermost conductive patterns **124**) on the redistribution structure **110** and the dies **130**, **140**, and electrically connect the redistribution structure **110** to the dies **130**, **140**.

[0015] Then, an underfill **150** may be formed between the dies **130**, **140** and the redistribution structure **110**, surrounding the die connectors **134**, **144**. The underfill **150** may be formed by a capillary flow process after the dies **130**, **140** are attached, or may be formed by a suitable deposition method before the dies **130**, **140** are attached.

[0016] After forming the underfill **150**, an encapsulant **152** is formed over the dies **130**, **140** and the underfill **150**. The encapsulant **152** may be a molding compound, epoxy, or the like, and may be applied by compression molding, transfer molding, or the like. The encapsulant **152** may be formed over the redistribution structure **110** such that the dies **130**, **140** are buried or covered. The encapsulant **152** is then cured.

[0017] Referring to FIG. **1C**, a plurality of under-ball metallurgy (UBM) patterns **160** and a plurality of conductive terminals **162** are sequentially formed on the conductive vias **114**. In some embodiments, the carrier **102** is de-bonded and is separated from the passivation layer **112**. In some embodiments, the de-bonding process includes projecting a light such as a laser light or an UV light on the de-bonding layer **104** (e.g., the LTHC release layer) so that the carrier **102** can be easily

removed along with the de-bonding layer **104**. During the de-bonding step, a tape (not shown) may be used to secure the structure before de-bonding the carrier **102** and the de-bonding layer **104**. After removing the carrier **102** and the de-bonding layer **104**, the UBM patterns **160** are formed on the conductive vias **114** respectively. The UBM patterns **160** may be formed for ball mount. The UBM patterns **160** include aluminum, copper, nickel, or an alloy thereof. The removal of the carrier **102** and the de-bonding layer **104** and/or formation of the UBM patterns **160** and the conductive terminals **162** may be performed while the encapsulant **152** is on a tape.

[0018] Then, the conductive terminals **162** are placed on the UBM patterns **160**. The conductive terminals **162** may be controlled collapse chip connection (C4) bumps, solder balls such as a ball grid array (BGA), metal pillars, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive terminals **162** include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or combinations thereof. In an embodiment in which the conductive terminals **162** are solder bumps, the conductive terminals **162** are formed by initially forming a layer of solder through various methods such as evaporation, electroplating, printing, solder transfer, ball placement, or the like. In this embodiment, once a layer of solder has been formed, a reflow is performed to shape the material into the desired bump shape.

[0019] Referring to FIG. 1D, the encapsulant **152** is thinned to expose surfaces of the dies **130**, **140**. The thinning may be accomplished by a CMP, a grinding process, or the like. After the thinning, surfaces (e.g., top surfaces) of the encapsulant **152** and dies **130**, **140** are level. In some embodiments, the redistribution structure **110** and the encapsulant **152** are singulated (not shown) by a singulation process, thereby forming a semiconductor package **100**. The singulation may be performed while the redistribution structure **110** is on a tape. Singulation is performed along scribe line regions. In some embodiments, the singulation process includes a sawing process, a laser process, or combinations thereof. As shown in FIG. 1D, as a result of the singulation process, sidewalls of the redistribution structure **110** and the encapsulant **152** are substantially flush with each other. The semiconductor package **100** may integrate homogeneous or heterogeneous components. In some embodiments, the semiconductor package **100** is formed by forming the redistribution structure first, which is also referred to a redistribution layer (RDL) first process. However, the disclosure is not limited thereto.

[0020] Referring to FIG. 1E, the semiconductor package **100** may be mounted onto a package substrate **202**, to form a semiconductor package **200**. In some embodiments, the package substrate **202** is an interposer such as an organic interposer. The package substrate **202** may include a redistribution structure. For example, the redistribution structure includes alternating dielectric layers (not shown) and conductive patterns (not shown) and a plurality of bond pads **204** over the dielectric layers and the conductive patterns. The conductive patterns may include vias and/or traces. The dielectric layers may include organic material, and the conductive patterns and the bond pads **204** may include conductive material such as copper. The conductive patterns and the bond pads **204** may be formed through any suitable process such as deposition, single damascene, dual damascene, or the like. In some embodiments, the package substrate **202** is substantially free of active and passive devices. However, the disclosure is not limited thereto.

[0021] In alternative embodiments, the package substrate **202** is made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations thereof, and the like, may also be used. Additionally, the package substrate **202** may be a SOI substrate. Generally, an SOI substrate includes a layer of a semiconductor material such as epitaxial silicon, germanium, silicon germanium, SOI, SGOI, or combinations thereof. The package substrate **202** may include active and passive devices (not shown). As one of ordinary skill in the art will recognize, a wide variety of devices such as transistors, capacitors, resistors, combinations of these,

and the like may be used to generate the structural and functional requirements of the design for the semiconductor package **200**. The devices may be formed using any suitable methods. The package substrate **202** may include the redistribution structure. The redistribution structure includes alternating dielectric layers (not shown) and conductive patterns (not shown) and a plurality of bond pads **204** over the dielectric layers and the conductive patterns. The conductive patterns and the bond pads **204** are formed over the active and passive devices and are designed to connect the various devices to form functional circuitry. The dielectric layers may include low-k dielectric material and the conductive patterns and the bond pads **204** may include conductive material such as copper. In alternative embodiments, the package substrate **202** is a printed circuit board such as a laminate substrate formed as a stack of multiple thin layers (or laminates) of a polymer material such as bismaleimide triazine (BT), FR-4, ABF, or the like. However, any other suitable substrate, such as an organic substrate, a ceramic substrate, or the like, may alternatively be utilized, and all such redistributive substrates that provide support and connectivity to the semiconductor package **100** are fully intended to be included in some embodiments.

[0022] In some embodiments, the conductive terminals **162** are reflowed to attach the semiconductor package **100** to the bond pads **204**, thereby bonding the semiconductor package **100** to the package substrate **202**. The conductive terminals **162** electrically and/or physically couple the package substrate **202** to the semiconductor package **100**. In alternative embodiments, passive devices (e.g., surface mount devices (SMDs), not illustrated) may be attached to the package substrate **202** (e.g., bonded to the bond pads **204**) prior to mounting the semiconductor package **100** on the package substrate **202**. In such embodiments, the passive devices may be bonded to a same surface of the package substrate **202** as the conductive terminals **162**.

[0023] In some embodiments, after bonding the semiconductor package **100** onto the package substrate **202**, an underfill **210** is formed between the semiconductor package **100** and the package substrate **202**, surrounding the conductive terminals **162**. The underfill **210** may be formed by a capillary flow process.

[0024] In some embodiments, the geometry and the arrangement of the conductive patterns **116** with respect to the die (e.g., the die **130**) play significant roles in ensuring the reliability of the formed semiconductor package **100**. The configuration of the conductive patterns **116** will be described in detail below in conjunction with FIG. 2A and FIG. 2B.

[0025] FIG. 2A and FIG. 2B are simplified top views illustrating various arrangements of the conductive patterns **116** in FIG. 1E. Referring to FIG. 1E, in some embodiments, the conductive patterns **116** are disposed directly under the die **130**. For example, the conductive patterns **116** are disposed under the die **130** within a periphery **130p** of the die **130**. The conductive patterns **116** may be arranged to surround a center axis CA of the die **130** (also referred to as an extending line of the center axis CA). Herein, the center axis is a line passing through a center of an object, and the object is substantially symmetric with respect to the center axis. The conductive patterns **116** may be arranged in an array. As illustrated in FIG. 2A, a top view of the conductive pattern **116** has an ellipse-like shape. In other words, a cross-section of the conductive pattern **116** substantially parallel to a surface **130a** of the die **130a** has an ellipse-like shape. The surface **130a** may be an active surface or a rear surface opposite to the active surface. Herein, the term “ellipse” refers to a symmetric shape having two mutually perpendicular axes which intersect at the center thereof due to its symmetry. Similarly, the term “ellipse-like” refers to a shape like an ellipse and having two mutually perpendicular axes which intersect at the center thereof. The ellipse-like shape may not be provided with a smooth periphery or contour and may not be as symmetric as an ellipse due to the process variation and tolerance. However, viewing the shape as the whole, the ellipse-like can be interpreted as an ellipse to some extent. The ellipse-like shape may be an ellipse (as shown in FIG. 4A and FIG. 4B), an oval (as shown in FIG. 2A and FIG. 2B), an oblong (as shown in FIG. 2A and FIG. 2B) or the like. The oblong is a shape having two long sides and two short sides, does not have right angles, and the short sides may be curved in some embodiments. As illustrated in FIG.

2A, the oval (the cross-section of the conductive patterns **116**) has a long-axis **118a** and a short-axis **118b** perpendicular to the long-axis **118a**. The long-axis **118a** of the conductive pattern **116** is aligned with the center axis CA of the die **130**. That is, the long-axis **118a** points toward the center axis CA of the die **130**, and the long-axis **118a** (also referred to as an extending line of the long-axis **118a**) intersects with the center axis CA of the die **130**. In some embodiments, the long-axis **118a** of each conductive pattern **116** intersects with the center axis CA of the die **130**.

[0026] A length L.sub.a of the long-axis **118a** is larger than a length L.sub.b of the short-axis **118b**. In other words, the conductive pattern **116** has a longer length L.sub.b. In some embodiments, a ratio of the length L.sub.a of the long-axis **118a** to the length L.sub.b of the short-axis **118b** is greater than 1 and less than 8. In some embodiments, the length L.sub.a of the long-axis **118a** ranges from 60 μm to 110 μm . The length L.sub.a of the long-axis **118a** may be larger than a maximum length of the UBM pattern **160**. For example, the UBM pattern **160** has a circular shape and has a diameter t1. In some embodiments, the length L.sub.a of the long-axis **118a** of the conductive pattern **116** is larger than the diameter t1 of the UBM pattern **160**, and the diameter t1 of the UBM pattern **160** is larger than or substantially equal to the length L.sub.b of the short-axis **118b** of the conductive pattern **116**. In alternative embodiments, the UBM pattern **160** is octagonal shaped. In some embodiments, a center axis CA1 of the conductive pattern **116** is substantially collinear with a center axis CA2 of the conductive via **114**. That is, the center axis CA1 is entirely overlapping with the center axis CA2. For example, a distance (i.e., the shortest distance) d1 between the center axis CA1 of the conductive pattern **116** to the center axis CA of the die **130** is the same as a distance (i.e., the shortest distance) d2 between the center axis CA2 of the conductive via **114** to the center axis CA of the die **130**. In some embodiments, the center axis CA1 of the conductive pattern **116** is substantially collinear with a center axis CA3 of the UBM pattern **160** and a center axis (not shown) of the conductive terminal **162**. However, the disclosure is not limited thereto. In some embodiments, the conductive patterns **116** are also referred to as cap patterns since the conductive patterns **116** cover tops of the conductive terminals **162**.

[0027] In some embodiments, the conductive patterns **116** are arranged to be tilted with respect to extending directions of sidewalls of the die **130**. For example, as illustrated in FIG. 2A, the die **130** is a parallelogram having four sides S1-S4 connected to each other. The side S1 and the side S3 may extend along a first direction D1 while the side S2 and the side S4 may extend along a second direction D2. The first direction D1 is perpendicular to the second direction D2, for example. In some embodiments, the long-axis **118a** of the oval (the cross-section of the conductive patterns **116**) forms an included angle $\theta_{\text{sub.a}}$ with the first direction D1, and the included angle $\theta_{\text{sub.a}}$ is larger than 0° and smaller than 90° . In some embodiments, the long-axis **118a** of the oval (the cross-section of the conductive patterns **116**) forms an included angle $\theta_{\text{sub.b}}$ with the second direction D2, and the included angle $\theta_{\text{sub.b}}$ is larger than 0° and smaller than 90° .

[0028] FIG. 2A also shows a projection of the die **130**, the conductive patterns **116**, the conductive vias **114** and the UBM patterns **160** onto a surface such as a front or rear surface of the die **130** and a front or rear surface of the package substrate **202**. For example, FIG. 2A illustrates a projection of the die **130**, the conductive patterns **116**, the conductive vias **114** and the UBM patterns **160** onto the surface **130a** of the die **130** or a surface **202a** of the package substrate **202**. As shown in FIG. 2A, in some embodiments, in the projection onto the surface **130a** or **202a**, a center C1 of the conductive pattern **116** is overlapped with a center C2 of the conductive via **114** and a center C3 of the UBM pattern **160**. The long-axis **118a** of the conductive pattern **116** is in a direction D from the center C1 of the conductive pattern **116** to a center C of the die **130**. In other words, the conductive pattern **116** is enlarged/extended along the direction D and thus has a larger length L.sub.a in the direction D.

[0029] In some embodiments, at the cooling stage, a coefficient of thermal expansion (CTE) mismatch is present between the package substrate **202** and the die **130**, which may induce stress concentration at an outer edge of the conductive pattern **116** and lead to a crack of the passivation

layer **112** aside the conductive pattern **116**. In some embodiments, by extending the conductive pattern **116** along the direction D from the center C1 of the conductive pattern **116** to the center C of the die **130**, the conductive pattern **116** has an ellipse-like shape, and the stress is reduced.

[0030] Referring to FIG. 2B, an alternative configuration of the conductive patterns **116** is provided. As illustrated in FIG. 2B, the conductive patterns **116** directly under the die **130** may be constituted by a plurality of first conductive patterns **116a** and a plurality of second conductive patterns **116b**. As illustrated in FIG. 2B, a top view of the first conductive pattern **116a** has an ellipse-like shape while a top view of the second conductive pattern **116b** has a circular shape. In other words, a cross-section of the first conductive pattern **116a** substantially parallel to the surface **130a** of the die **130** has an ellipse-like shape, and a cross-section of the second conductive pattern **116b** parallel to the surface **130a** of the die **130** has a circular shape. The first conductive patterns **116a** and the UBM patterns **160** are similar to the conductive patterns **116** and the UBM patterns **160** of FIG. 2A except that the first conductive patterns **116a** are arranged at regions corresponding to corners **130c** of the die **130**. For example, as illustrated in FIG. 2B, the second conductive patterns **116b** are sandwiched between the first conductive patterns **116a**. The long-axis **118a** of the first conductive pattern **116a** intersects with the center axis CA of the die **130**. In some embodiments, the long-axis **118a** of each conductive pattern **116a** intersects with the center axis CA of the die **130**. In some embodiments, the length L.sub.a of the long-axis **118a** of the first conductive pattern **116a** is larger than the diameter t1 of the UBM pattern **160**, and the diameter t1 of the UBM pattern **160** is larger than or substantially equal to the length L.sub.b of the short-axis **118b** of the first conductive pattern **116a**. A diameter t2 of the second conductive pattern **116b** is smaller than the diameter t1 of the UBM pattern **160**, for example. In some embodiments, the center axis CA1 of the first conductive pattern **116a** is substantially collinear with the center axis CA2 of the conductive via **114**. In some embodiments, the center axis CA1 of the first conductive pattern **116a** is substantially collinear with the center axis CA3 of the UBM pattern **160** and the center axis (not shown) of the conductive terminal **162**. Similarly, in some embodiments, the center axes (not shown) of the second conductive pattern **116b**, the conductive via **114**, the UBM pattern **160** and the conductive terminal **162** are substantially collinear with one another. However, the disclosure is not limited thereto. In addition, in alternative embodiments, the second conductive patterns **116b** disposed between the first conductive patterns **116a** have other suitable shape. In an embodiment in which the second conductive pattern **116b** has the same shape as the first conductive pattern **116a**, the long-axis of the second conductive pattern **116b** may or may not intersect with the center axis CA of the die **130**.

[0031] Similar to FIG. 2A, FIG. 2B also shows a projection of the die **130**, the conductive patterns **116**, the conductive vias **114** and the UBM patterns **160** onto the surface **130a** of the die **130** or the surface **202a** of the package substrate **202**. As shown in FIG. 2B, in some embodiments, in the projection onto the surface **130a** or **202a**, a center C1 of the conductive pattern **116a** is overlapped with a center C2 of the conductive via **114** and a center C3 of the UBM pattern **160**. Similarly, centers (not shown) of the conductive pattern **116b**, the conductive via **114** and the UBM pattern **160** are overlapped. The long-axis **118a** of the conductive pattern **116a** is in a direction D from the center C1 of the conductive pattern **116a** to a center C of the die **130**. In other words, the conductive pattern **116a** is enlarged/extended along the direction D and thus has a larger length L.sub.a in the direction D.

[0032] In some embodiments, since the stress mentioned above is larger at regions corresponding to the corners **130c** of the die **130** than other sites, the conductive patterns **116a** at these regions are designed to have an ellipse-like shape. In other words, the conductive patterns **116a** in the regions corresponding to the corners **130c** of the die **130** are configured to have a larger size than the UBM patterns **160** therebeneath while the conductive patterns **116b** in other regions may remain to have a size smaller than the UBM patterns **160** therebeneath.

[0033] In alternative embodiments, as shown in FIGS. 3A and 3B, the configuration of the

conductive structures **200** is similar to that of FIG. **1E** and FIG. **2A**, and the main difference lies the center axis **CA2** of the conductive via **114** is not collinear with the center axis **CA1** of the conductive pattern **116**. For example, as shown in FIGS. **3A** and **3B**, the center axis **CA2** of the conductive via **114** is shifted with respect to the center axis **CA1** of the conductive pattern **116**, so as to be farther away from the center axis **CA** of the die **130** than the center axis **CA1** of the conductive pattern **116**. In some embodiments, the center axis (not shown) of the conductive terminal **162** is collinear with the center axis **CA1** of the conductive pattern **116**. In some embodiments, a distance **d1** (i.e., the shortest distance) between the center axis **CA1** of the conductive pattern **116** to the center axis **CA** of the die **130** is smaller than a distance **d2** (i.e., the shortest distance) between the center axis **CA2** of the conductive via **114** to the center axis **CA** of the die **130**. In some embodiments, a distance **d3** between the center axis **CA2** of the conductive via **114** and the center axis **CA1** of the conductive pattern **116** ranges from 3 μm to 30 μm .

[0034] FIG. **3B** also shows a projection of the die **130**, the conductive patterns **116**, the conductive vias **114** and the UBM patterns **160** onto a surface such as a front or rear surface of the die **130** and a front or rear surface of the package substrate **202**. For example, FIG. **3B** illustrates a projection of the die **130**, the conductive patterns **116**, the conductive vias **114** and the UBM patterns **160** onto the surface **130a** of the die **130** or the surface **202a** of the package substrate **202**. As shown in FIG. **3B**, in some embodiments, in the projection onto the surface **130a** or **202a**, the center **C1** of the conductive pattern **116** is overlapped with the center **C3** of the UBM pattern **160**. In some embodiments, the center **C2** of the conductive via **114** is shifted with respect to the centers **C1**, **C3** of the conductive pattern **116** and the UBM pattern **160**. The center **C2** of the conductive via **114** may be shifted outward along the direction **D** from the center **C1** of the conductive pattern **116** to the center **C** of the die **130** with respect to the center **C1** of the conductive pattern **116**. A distance **d4** between the center **C2** of the conductive via **114** to an inner edge of the UBM pattern **160** along the direction **D** is larger than a distance **d5** between the center **C2** of the conductive via **114** to an outer edge of the UBM pattern **160** along the direction **D**. A difference between the distance **d4** and the distance **d5** is equal to the distance **d3** and ranges from 3 μm to 30 μm . The conductive pattern **116** has a larger length **L.sub.a** in the direction **D** from the center **C1** of the conductive pattern **116** to the center **C** of the die **130**. In other words, the conductive pattern **116** is enlarged along the direction **D**. In some embodiments, by shifting the conductive vias **114** outward along the direction **D** from the center **C1** of the conductive pattern **116** to the center **C** of the die **130**, the stress is further reduced.

[0035] In above embodiments, the conductive patterns **116** disposed directly under the die **130** are configured as FIG. **2A**, FIG. **2B**, FIG. **3B** or other similar arrangement, and the conductive patterns **116** disposed directly under the die **140** may have other arrangement. For example, each of the conductive patterns **116** disposed directly under the die **140** has a circular shape, and each of these conductive patterns **116** has a smaller diameter than the UBM patterns **160** or substantially the same diameter as the UBM patterns **160**. However, the disclosure is not limited thereto. In alternative embodiments, the conductive patterns **116** disposed directly under the die **140** are also configured as the conductive patterns **116** of FIG. **2A**, FIG. **2B** and FIG. **3B** or other similar arrangement. In other words, according to the design, the conductive patterns **116** and/or the conductive vias **114** may be configured corresponding to one or more dies of the semiconductor package.

[0036] In some embodiments, by extending the conductive pattern of the redistribution structure along the direction toward to the center axis of the die, the conductive pattern has an ellipse-like shape. In addition, in some embodiments, the conductive via is shifted outward along the direction with respect to the center of the conductive pattern. Accordingly, the stress due to the CTE mismatch between the package substrate and the die may be reduced. Therefore, the performance of the formed semiconductor package may be improved.

[0037] In accordance with some embodiments of the disclosure, a semiconductor package includes

a die, a redistribution structure and a plurality of conductive terminals. The redistribution structure is disposed below and electrically connected to the die. The redistribution structure includes a plurality of conductive patterns, and at least one of the plurality of conductive patterns has a cross-section substantially parallel to the surface of the die. The cross-section has a long-axis and a short-axis, and the long-axis intersects with a center axis of the die. The conductive terminals are disposed below and electrically connected to the redistribution structure.

[0038] In accordance with some embodiments of the disclosure, a semiconductor package includes a redistribution structure, a die, an underfill and an encapsulant. The redistribution structure includes a plurality of first conductive patterns and a plurality of second conductive patterns. The plurality of first conductive patterns have an ellipse-like shape respectively, and the plurality of second conductive patterns have a circular shape respectively. The die is disposed over and electrically connected to the redistribution structure. The underfill is disposed between the die and the redistribution structure. The encapsulant encapsulates the die over the redistribution structure.

[0039] In accordance with some embodiments of the disclosure, a semiconductor package includes a die, a redistribution structure and a plurality of under-ball metallurgy (UBM) patterns. The die has a first center axis. The redistribution structure is disposed below and electrically connected to the die, and includes a plurality of conductive patterns and a plurality of conductive vias. The plurality of conductive vias are disposed between and in direct with the plurality of conductive patterns and the plurality of UBM patterns. One of the plurality of conductive vias has a second center axis, a respective one of the plurality of UBM patterns under the one of the plurality of conductive vias has a third center axis, and a distance between the second center axis and the first center axis is larger than a distance between the third center axis and the first center axis.

[0040] In accordance with some embodiments of the disclosure, a semiconductor package includes a die, a first conductive pattern, a second conductive pattern and first and second under-ball metallurgy (UBM) patterns. The first conductive pattern and the second conductive pattern are disposed below and electrically connected to the die, wherein the first conductive pattern has an ellipse-like shape, and the second conductive pattern has a circular shape. The first and second UBM patterns correspond to the first and second conductive patterns, the first conductive pattern has a first length, the second conductive pattern has a second length, the first and second UBM patterns have a third length, wherein the first length is larger than the third length and the second length is smaller than the third length.

[0041] In accordance with some embodiments of the disclosure, a semiconductor package includes a die and a redistribution structure. The redistribution structure is electrically connected to the die. The redistribution structure includes a plurality of first conductive patterns and a plurality of second conductive patterns, wherein the first conductive patterns have an ellipse-like shape respectively, the second conductive patterns have a circular shape respectively, and the first conductive patterns are arranged at corners of the die.

[0042] In accordance with some embodiments of the disclosure, a semiconductor package includes a die, a conductive pattern, a conductive via and an under-ball metallurgy (UBM) pattern. The die has a first center axis. The conductive via is disposed between the conductive pattern and the UBM pattern, wherein the conductive via has a second center axis, the UBM pattern has a third center axis, and a distance between the second center axis and the first center axis is larger than a distance between the third center axis and the first center axis.

[0043] In accordance with some embodiments of the disclosure, a semiconductor package includes a die and a redistribution structure. The die has a plurality of corner regions and a first region between the corner regions. The redistribution structure is electrically connected to the die and includes a plurality of first conductive patterns disposed in the corner regions and a plurality of second conductive patterns disposed in the first region. The plurality of first conductive patterns have an ellipse-like shape respectively, and the plurality of second conductive patterns have a circular shape respectively.

[0044] In accordance with some embodiments of the disclosure, a semiconductor package includes a die, a conductive pattern, a conductive via and an under-ball metallurgy (UBM) pattern. The die has a first center axis. The conductive via is disposed between the conductive pattern and the UBM pattern, wherein the conductive via has a second center axis, the UBM pattern has a third center axis, a distance between the second center axis and the first center axis is larger than a distance between the third center axis and the first center axis, and the conductive via is in direct contact with the conductive pattern and the UBM pattern.

[0045] In accordance with some embodiments of the disclosure, a semiconductor package includes a conductive pattern, a conductive via and an under-ball metallurgy (UBM) pattern. The conductive pattern has an ellipse-like shape. The conductive via is disposed between and in direct contact with the conductive pattern and the UBM pattern.

[0046] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the disclosure. Those skilled in the art should appreciate that they may readily use the disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the disclosure.

Claims

1. A semiconductor package, comprising: a die, having a plurality of corner regions and a first region between the corner regions; and a redistribution structure electrically connected to the die, comprising a plurality of first conductive patterns disposed in the corner regions and a plurality of second conductive patterns disposed in the first region, wherein the plurality of first conductive patterns have an ellipse-like shape respectively, and the plurality of second conductive patterns have a circular shape respectively.
2. The semiconductor package according to claim 1, wherein the first region is cross-shaped.
3. The semiconductor package according to claim 1, wherein in a top view, centers of some of the first conductive patterns and centers of some of the second conductive patterns are arranged along a direction.
4. The semiconductor package according to claim 3, wherein two opposite sides of the die are arranged along a first direction and another two opposite sides of the die are arranged along a second direction perpendicular to the first direction, the direction forms an included angle with the first direction, and the included angle is larger than 0° and smaller than 90° .
5. The semiconductor package according to claim 1, further comprising a first conductive via between one of the first conductive patterns and a first under-ball metallurgy (UBM) pattern and a second conductive via between one of the second conductive patterns and a second UBM.
6. The semiconductor package according to claim 5, wherein in a top view, centers of the one of the first conductive patterns, the first conductive via and the first UBM pattern are overlapped.
7. The semiconductor package according to claim 5, wherein the first conductive via is in direct contact with the one of the first conductive patterns and the first UBM pattern.
8. A semiconductor package, comprising: a die, having a first center axis; a conductive pattern and a conductive via; and an under-ball metallurgy (UBM) pattern, the conductive via disposed between the conductive pattern and the UBM pattern, wherein the conductive via has a second center axis, the UBM pattern has a third center axis, a distance between the second center axis and the first center axis is larger than a distance between the third center axis and the first center axis, and the conductive via is in direct contact with the conductive pattern and the UBM pattern.
9. The semiconductor package according to claim 8, wherein a center axis of the conductive pattern is overlapped with the third center axis of the UBM pattern.

- 10.** The semiconductor package according to claim 8, wherein in a top view, a center of the conductive via, a center of the conductive pattern and a center of the die are arranged along a direction.
- 11.** The semiconductor package according to claim 8, wherein in a top view, the conductive pattern has an ellipse-like shape.
- 12.** The semiconductor package according to claim 11, wherein two opposite sides of the die are arranged along a first direction and another two opposite sides of the die are arranged along a second direction perpendicular to the first direction, a long-axis of each of the plurality of conductive patterns forms an included angle with the first direction, and the included angle is larger than 0° and smaller than 90° .
- 13.** The semiconductor package according to claim 8, wherein in a top view, the conductive via and the UBM pattern respectively have a circular shape.
- 14.** A semiconductor package, comprising: a conductive pattern and a conductive via, wherein the conductive pattern has an ellipse-like shape; and an under-ball metallurgy (UBM) pattern, wherein the conductive via is disposed between and in direct contact with the conductive pattern and the UBM pattern.
- 15.** The semiconductor package according to claim 14, wherein in a top view, a center of the conductive pattern and a center of the UBM pattern are overlapped.
- 16.** The semiconductor package according to claim 15, wherein a center of the conductive via is further overlapped with the centers of the conductive pattern and the UBM pattern.
- 17.** The semiconductor package according to claim 14, wherein in a top view, the conductive via and the UBM pattern respectively have a circular shape.
- 18.** The semiconductor package according to claim 14, wherein the conductive pattern has a first length, the conductive via has a second length, the UBM pattern has a third length, and the first length is larger than the third length and the second length is smaller than the third length.
- 19.** The semiconductor package according to claim 18, wherein the conductive pattern has the first length along a first direction and a fourth length along a second direction, and the fourth length is between the second length and the third length.
- 20.** The semiconductor package according to claim 19, further comprising a die disposed over and electrically connected to the conductive pattern, wherein two opposite sides of the die are arranged along a third direction and another two opposite sides of the die are arranged along a fourth direction perpendicular to the third direction, the first direction forms an included angle with the third direction, and the included angle is larger than 0° and smaller than 90° .
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